

and maximum control. Django, with its comprehensive features and conventions, is ideal for large scale projects and applications with complex requirements.

Key features

- *Batteries-included philosophy:* A rich set of built-in features out-of-the-box, including an ORM (object relational mapper), admin interface, URL routing, authentication, and security features, enabling rapid development of fully-featured web applications.
- *Model-view-template (MVT) architecture.*
- *Admin interface:* Django provides a customisable admin interface for managing application data, offering CRUD functionality and the ability to customise views, filters, and permissions, streamlining the development of administrative interfaces for web applications.

Limitations

- *Performance overhead:* High-level abstractions and ORM in Django may introduce performance overhead compared to lower-level frameworks or custom solutions optimised for specific use cases.

Gradio: Intuitive and interactive machine learning interface

Gradio is a Python library that simplifies the creation and deployment of interactive machine learning models and applications. It provides a user-friendly interface for customisable input and output interfaces, enabling users to interact with models through web interfaces without coding. This accessibility caters to a wide range of users, including machine learning experts, domain specialists, and non-technical users.

Used in rapid prototyping, education, research, and industry, Gradio facilitates quick model prototyping, interactive teaching

demonstrations, collaborative model showcases, and the development of interactive dashboards and applications for data analysis, decision support, and customer engagement.

Key features

- Gradio provides a simple and intuitive interface for building interactive machine learning applications without writing code. Its user-friendly design allows developers to quickly prototype and deploy machine learning models with minimal effort, making it accessible to users with varying levels of technical expertise.
- Gradio allows developers to create customisable input and output interfaces for machine learning models, including text inputs, image inputs, sliders, checkboxes, and visualisations. This flexibility enables developers to tailor the user experience and visualise model predictions in real-time, enhancing interactivity and user engagement.

Limitations

- Limited scalability
- Limited support for complex models

Tkinter: Pythonic GUI toolkit

Tkinter is a versatile and powerful Python GUI development tool, offering a wide range of widgets and tools for creating modern, user-friendly applications. It provides components from basic buttons to complex widgets like Canvas and Treeview. Tkinter's cross-platform compatibility allows developers to create applications that run seamlessly on Windows, MacOS, and Linux, making it a top choice for broad audience reach without platform-specific development. Its simplicity and ease of learning make it an excellent choice for developers of all levels. While its default look may seem outdated, Tkinter offers themes and styling options for customising

application appearances. Overall, Tkinter is a reliable and powerful tool for Python GUI development.

Key features

- Simple and lightweight GUI toolkit that comes bundled with Python.
- Cross-platform compatibility to run seamlessly on various operating systems, including Windows, MacOS, and Linux, without modification.
- Rich widget set.

Limitations

- Limited styling and customisation
- Limited layout management

Pandas: Flexible data manipulation library

Pandas is a core tool for data processing, analysis and research in Python. Its intuitive API and extensive functionality make it popular among data scientists and analysts. Whether cleaning messy data, performing complex transformations, or conducting statistical analysis, Pandas simplifies manipulation of various datasets.

Built on NumPy, Pandas offers DataFrames and Series. DataFrames are two-dimensional labelled data structures similar to database tables, while Series are one-dimensional labelled arrays similar to spreadsheet columns. These structures enable efficient data processing, offering functions for cleaning, transforming, and manipulating data. Pandas allows operations like filtering, sorting, grouping, combining, and editing data, facilitating extraction of insights from complex datasets.

Key features

- *Powerful data structures:* Pandas provides highly flexible and efficient data structures, primarily Series (one-dimensional) and DataFrame (two-dimensional), which are capable of handling labelled and structured data effectively.